

Shamba Pulse

Late-blight warnings and spray timing for smallholder tomato and potato farmers around Juja, Kiambu — built on real data from the JKUAT Conduit weather station.

Hack The Weather 2026

JHUB Africa / JKUAT

Team: [Team name]

Live app: shamba-pulse-jkuat.streamlit.app

github.com/Hackathons-4thyear/HackTheWeather

The problem, and who it is for

Late blight (*Phytophthora infestans*) can destroy a tomato or potato crop in under a week. The wet, humid spells of an El Niño short-rains season are exactly when it spreads — and exactly when a farmer can least afford to guess. Two guesses cost money: spraying on a fixed schedule buys fungicide that may not be needed, and spraying at the wrong hour wastes it anyway, because rain washes it off, dew dilutes it and still air lets it drift.

Farmers — the product is the SMS

A smallholder does not open a dashboard during planting season. They get one message, under 160 characters, in English or Kiswahili, on any phone, and only when the risk has actually changed.

Extension officers, agrovets, cooperatives

They advise many farmers and need the reasoning: the forecast, the spray windows and the evidence trail behind every alert. One of them serves hundreds of farms, which is what makes a single weather station worth building on.

How it works

Blight risk — the Hutton criteria

A **Hutton day** needs **both** a daily minimum temperature of at least **10 °C** and **six or more hours** at **90% relative humidity** or above. **Two consecutive Hutton days is a HIGH-risk warning** — the operational standard from the James Hutton Institute.

HIGH is reserved for that trigger and nothing else. A secondary humid-hours measure grades the last 24 hours so there is always an answer mid-day or across gaps, but it can never produce HIGH on its own.

A day missing more than **25%** of its hours is marked *insufficient data* and is not judged; it **breaks** a consecutive run rather than being bridged, because a missing night is exactly when the humid hours would have happened.

Send policy — what actually gets texted

LOW/UNKNOWN never text. **HIGH** is always eligible, and an escalation into HIGH ignores the cooldown entirely. **MODERATE** texts only on escalation, never while it persists. A 3-day cooldown applies per level. The dashboard still shows every day's level — the policy governs SMS only.

Spray windows

An hour is sprayable only when all of these hold:

- not raining now (≤ 0.2 mm, a gauge's resolution)
- dry for **6 hours** afterwards, so contact fungicide becomes rain-fast
- wind **1–4 m/s** — below drifts, above blows off target
- humidity **under 90%**, so dew does not dilute the spray
- **daylight only** (06:30–18:30)
- at least **2 hours** long after clipping

Only the spraying is daylight-limited — the 6-hour dry check looks through the night, because rain at 2am still washes off a 7pm spray. Every threshold sits in one `config.py` with a comment explaining the number.

Kiswahili is domain knowledge, not translation

Times use **Swahili time**, counted from dawn: 09:00 is *saa 3 asubuhi*, 17:00 is *saa 11 jioni*. Because "saa 11" means 17:00, a duration of 11 hours is written *masaa 11* so the two cannot be confused in one sentence. Late blight is named fully as *baka chelewa*.

Use of the JKUAT Conduit station data

Readings come from a POST to `conduit.jhubafrica.com/data.php` (form-encoded: `apikey`, `email`, `fromdate`, `todate`). We downloaded the full archive in cached 7-day chunks at one request per second.

476

days of station archive

46,183

readings, one every 15 min

98.6–99.1%

monthly coverage in OND 2025

0

gaps longer than 6 hours

We verified the sensors instead of trusting them

Four faults, each found by looking at the data. Three of them would have produced confidently wrong farmer advice.

Finding	Evidence	What we did
Rainfall field unusable	rg1 recorded 18.6 mm against 301.8 mm actual over OND 2025 — it captures 6.2% of rainfall.	Rainfall rebuilt by differencing the running daily total <code>rg1tt</code> , treating the drop to zero as the station's UTC-midnight reset.
Rain gauge 2 faulty	rg2tt resets about 21 times a day instead of once, reconstructing to 4,365 mm in a ~300 mm season. rg2 is constant zero.	Never used, and specifically never used to fill gaps in gauge 1 — which an earlier version of our code did.
Timestamps are UTC	The API returns <code>2026-09-20T00:00:26Z</code> ; Kenya is UTC+3 .	Converted to Africa/Nairobi on ingest. Read naively, every overnight humid-hours count shifts three hours and the daily boundary moves — silently.
Two dead / duplicated fields	<code>si1145_uv</code> is constant zero across all readings; <code>wind_gust_dir</code> is identical to <code>wind_gust</code> on 100% of rows and ranges -999.9 to 13.9.	Neither is used. Light channels are labelled raw counts, never W/m^2 .

A broken sensor is more dangerous than a missing one, because it looks like data. We have drafted a note to the JHUB team about gauge 2 rather than quietly working around it.

Does it work? The OND 2025 backtest

We replayed the real October–December 2025 short rains through the engine, one morning at a time, with the engine seeing only what had happened by that point. **No lookahead** — enforced in code and asserted by tests.

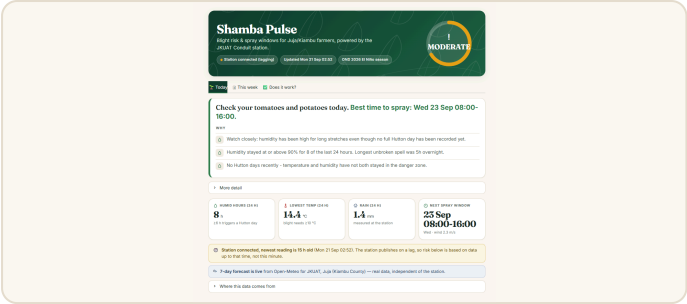
Result over 92 days (8,737 real observations)	Value
Days at HIGH blight risk	10
Distinct HIGH episodes	6
Longest Hutton run	5 consecutive days
Days inside a Hutton run	21
SMS if we texted every eligible day	46
SMS actually sent under the send policy	15 (67% fewer)
Days we refused to judge (data gaps)	0

Independent corroboration. The season's main event — 29 October to 1 November 2025, four straight days at HIGH — **coincided with** the Kenya Meteorological Department's heavy-rainfall advisory for 23–30 October, which named Kiambu and was expected to mark the onset of the short rains. Our engine flagged it from station humidity and temperature alone, with no knowledge of the advisory. That corroborates the weather was genuinely unusual. **It is not evidence that blight occurred** — nobody surveyed the fields.

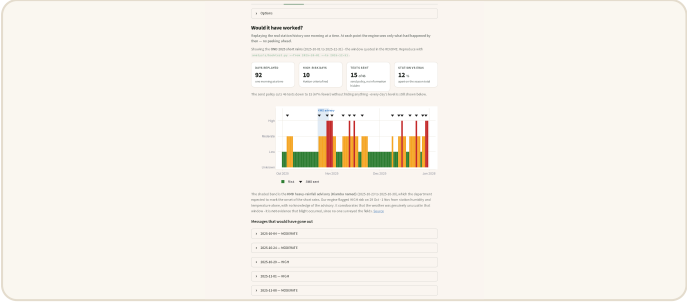
ERA5 cross-check on our rebuilt rainfall

Because rainfall is reconstructed rather than read directly, we compared it with ERA5 for the same coordinates over the same 92 days: **301.8 mm (station) against 268.6 mm (ERA5) — within 12%**. ERA5 is a **reanalysis**, a model reconstruction that assimilates satellite and ground observations onto a ~9 km grid, **not a direct satellite measurement**. The station is one point inside such a cell and rainfall here is convective, so the weak day-to-day correlation is the expected result while the seasonal agreement is the meaningful one. This is a credibility check, **not calibration** — no station value is adjusted toward ERA5.

The dashboard



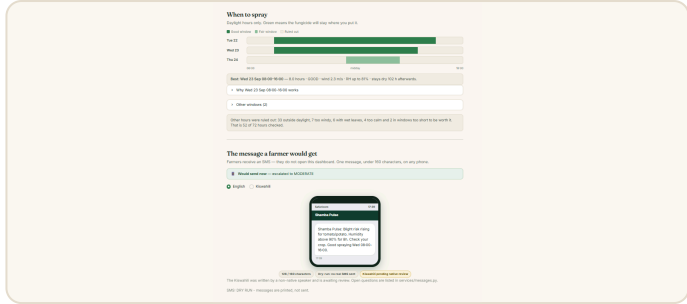
The verdict first: a risk ring stating the level in words as well as colour, one imperative sentence naming the actual spray window, the reasons behind it, and the numbers that drove it. The “lagging” pill is the app reporting that the station's newest reading is hours old rather than calling stale data “live”.



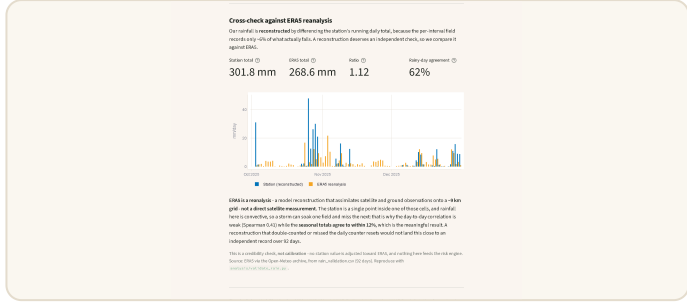
Every day of the OND 2025 short rains replayed under the headline numbers. Red is HIGH risk, triangles are messages that would have been sent, and the shaded band is the KMD advisory.

Limitations

- **One station.** Every reading comes from a single point at JKUAT. Convective rain soaks one field and misses the next.
- **No soil-moisture or leaf-wetness sensor exists** on this station and we never fabricate one. Humidity $\geq 90\%$ is our stated leaf-wetness proxy.
- **The Hutton criteria were developed in the UK** and encode a temperate understanding of the pathogen. Local strains, cultivars and highland conditions may shift the thresholds.



Spray windows drawn across each day's daylight hours, with a plain sentence saying what was ruled out and why — then the exact SMS a farmer receives, shown on a phone, switchable between English and Kiswahili.



Our reconstructed rainfall against ERA5 reanalysis for the same season — 301.8 mm against 268.6 mm.

Honesty notes, stated plainly. (1) SMS runs in **dry-run mode** — no real message is sent. The Africa's Talking sender is implemented and works against their sandbox; dry-run is the default so a public demo cannot spend credit or text a real farmer. (2) The **Kiswahili is pending native review**. It was written by a non-native speaker; the app badges this on screen and the open questions are listed in the source. (3) Our impact figures are an **illustrative scenario, not a measured result** — 13 weekly scheduled sprays against 6 episode-targeted ones over the same 92 days. We publish **no yield or income numbers**, because we have not run a field trial and would be making them up.

What's next

1. **A farmer pilot with a county extension office** — put the SMS in front of real smallholders through the people who already advise them.
2. **Calibrate against observed outbreaks in Kiambu.** Pair extension-officer or farmer-reported blight records with our risk history and tune the thresholds to local conditions. Everything else is secondary.
3. **CHIRPS / GPM satellite rainfall in the live pipeline.** Today satellite-derived data appears only as an offline ERA5 sanity check.
4. **Multi-station coverage**, so risk can be interpolated across a farming area rather than spoken for from one point.
5. **An ML model trained on observed outbreaks.** `disease_engine.assess()` is a documented swap point: same input frame, same output contract. It needs labels first.
6. **Native Kiswahili review**, then flip the flag the app uses to badge it; and **USSD**, so a farmer can ask rather than wait, with no smartphone and no data bundle.

Built with Python 3.12 · pandas · numpy · Streamlit · Plotly · pyarrow · requests · Africa's Talking · Open-Meteo (forecast + ERA5 archive) · JKUAT Conduit weather station · Hutton criteria · pytest (378 tests). Rule-based and explainable throughout — every threshold is one commented line in `config.py`.